

Response to Reviewer Comments

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A New IoT Platform Based on PQC Algorithms: A Solution for Quantum Computing Threats

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We thank the reviewer for the careful assessment and for recognizing the practical, architecture-driven treatment of post-quantum cryptography (PQC) migration in heterogeneous Internet of Things (IoT) systems. We agree that the figures in the submitted manuscript did not meet publication quality. The four submitted figures have been consolidated into two author-redrawn vector schematics. Their SVG sources are retained with the manuscript, and cropped vector-PDF derivatives are used in LaTeX so that labels and line art remain sharp at publication scale. The title, scope, author list, algorithm-placement model, experimental data, and principal conclusions remain unchanged.

Response to the overall assessment

Reviewer assessment: *Borderline Accept. The paper's core strength lies in its practical, architecture-driven approach to migrating heterogeneous IoT ecosystems to PQC. The study introduces a strategic algorithm-placement model across device tiers.*

Response. We appreciate this assessment. The revised manuscript retains the architecture-driven scope and makes the placement argument easier to verify. Section III-A now presents the node, gateway, and server responsibilities as an explicit tiered placement model, and Table I states the placement and design rationale for ML-KEM, ML-DSA, SLH-DSA, and hybrid mode. The Introduction states the contribution in the same terms, while the Conclusion limits the claim to a migration-oriented architecture supported by the reported prototype measurements.

Response to the figure-quality comment

Reviewer comment: *I am mainly concerned about the AI-generated figures (Fig. 1 and Fig. 4). The human-generated figures (Fig. 2 and Fig. 3) are of low quality and therefore must be improved.*

Response. We agree. The submitted artwork has not been reused. The four-figure structure has been consolidated into two author-redrawn vector schematics to remove overlap among the overview, hardware, sequence, and deployment views. Both delivered figures consist of vector paths without embedded raster images. We also standardized line weights, typography, arrow directions, component boundaries, and captions so that the architecture remains legible in the IEEE two-column layout.

The consolidation was deliberate:

- **Fig. 1, “End-node architecture for PQC-enabled IoT devices,”** shows the processing core, PQC software stack, bootloader and secure boot, RAM, storage, network path, system bus, and peripheral interfaces discussed in Section III-C.
- **Fig. 2, “Three-tier deployment map for quantum-safe session establishment and passkey authentication,”** shows constrained and Linux-capable end nodes, gateway-side ML-KEM/KDF/AEAD and signature functions, optional direct PQC operation, and cloud/server trust services discussed in Section III-D.

The sequence previously shown in submitted Fig. 3 is now described in Section III-D rather than repeated as a separate diagram. The revised Fig. 2 incorporates the tier-placement content previously distributed across the overview and deployment figures. The manuscript now contains exactly two figures, with no obsolete references to Figs. 3 or 4. This revision changes the presentation, not the architecture or experimental claims.

Response to technical quality and soundness

Reviewer assessment: *Fair (Some technical issues or missing baselines, but potentially fixable).*

Response. We acknowledge the absence of a matched physical-board or accelerator baseline. No additional board-level experiment was available for this revision, so we did not introduce an unmeasured comparison. Instead, the revised manuscript makes the evaluation boundary explicit in Section IV: the reported values are software-centric QEMU prototype measurements used as placement evidence, not cycle-accurate hardware characterization. Section IV-C also states that Linux/QEMU resident-set size is not an ESP32 SRAM-footprint measurement. The limitations paragraph identifies physical-board power, radio throughput, flash wear, timing leakage, and comparisons with existing PQC IoT architectures as future work.

Within that stated scope, the technical presentation has been tightened in three places. Table I connects each primitive to a deployment tier and rationale. Table II reports the measured elapsed time, resident memory, payload size, and recommended tier without changing the underlying values. Section IV-D links the observations to buffer sizing, gateway admission control, and candidate acceleration kernels. The end-node description now distinguishes AXI/AHB/APB bus interfaces from memory-mapped I/O, and the secure-channel text identifies AES-GCM and ChaCha20-Poly1305 as example authenticated-encryption schemes. These revisions clarify what the evidence supports without extending the conclusions beyond the prototype.

Response to originality and novelty

Reviewer assessment: *Fair (Incremental improvement or a minor new application).*

Response. We have not reframed the work as a new cryptographic primitive. The revised Introduction states the contribution more narrowly: (1) a layered node/gateway/server architecture for PQC-enabled IoT, (2) a PQC-aware passkey metadata and hybrid-policy model, and (3) a measurement-driven rule for placing ML-KEM, ML-DSA, and SLH-DSA. Section III and the two revised figures are organized around these system-level contributions. This wording separates the paper’s architectural contribution from the standardized algorithms and software implementations on which the prototype relies.

Summary of manuscript changes

Reviewer concern	Revision	Location
AI-generated appearance in submitted Figs. 1 and 4	Replaced the submitted artwork with two author-redrawn SVG schematics and vector-PDF derivatives	Revised Figs. 1 and 2; Sections III-C–D
Low quality of submitted Figs. 2 and 3	Standardized line art, typography, arrows, labels, and captions; removed redundant figure content	Revised Figs. 1 and 2
Technical scope and possible missing baselines	Preserved the measured values, clarified the QEMU/prototype boundary and interface terminology, and identified unperformed board-level comparisons as future work	Section IV-A–D; Conclusion
Incremental novelty	Stated the contribution as tiered placement, migration metadata/policy, and measurement-guided system design rather than a new primitive	Introduction; Section III-A; Conclusion

We thank the reviewer again. The camera-ready manuscript now contains the final vector artwork and the corresponding figure references. No new experiment, numerical result, or cryptographic claim was introduced during the figure revision.